
SOFTWARE REQUIREMENTS SPECIFICATION

for

FOCUS FLOW

Version 1.0

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Revision History

Name	Date	Reason For Changes	Version
Focus Flow	15/01/2026	Initial SRS Documentation	1.0

1 Introduction

1.1 Purpose

Existing online examination and learning management systems (LMS) suffer from a significant lack of integrity and flexibility. Traditional platforms often fail to provide adequate safeguards against academic dishonesty during remote assessments. Furthermore, current systems lack integrated support for technical assessments, forcing institutions to use separate platforms for MCQs and practical coding tests. "Focus Flow" is the solution. It is a robust, unified web application that combines a secure exam environment with lightweight, client-side AI proctoring and an integrated coding sandbox. Refer Figure 1.1

1.2 Intended Audience and Reading Suggestions

This SRS is for developers, project guides, educational stakeholders, and testers. Further discussion will provide all the internal, external, functional, and non-functional information about "Focus Flow".

1.3 Project Scope

"Focus Flow" creates a space for Instructors, Students, and Educational Institutions to maintain academic standards during remote learning.

The application bridges the gap between assessment integrity and user experience. It allows specific roles:

- **Instructors:** Can create dynamic exams, manage question banks, and review flagged events from proctored sessions.
- **Students:** Can take exams in a secure environment, write code in an integrated sandbox, and view performance analytics.

The core scope includes User Registration, Exam Builder Dashboard, Integrated Coding Sandbox, AI-Based Proctoring, and Student Analytics.

Figure 1.1 represents the high-level workflow where the Student interacts with the Exam Interface, monitored locally by the AI module, while the Instructor manages the content and reviews results.

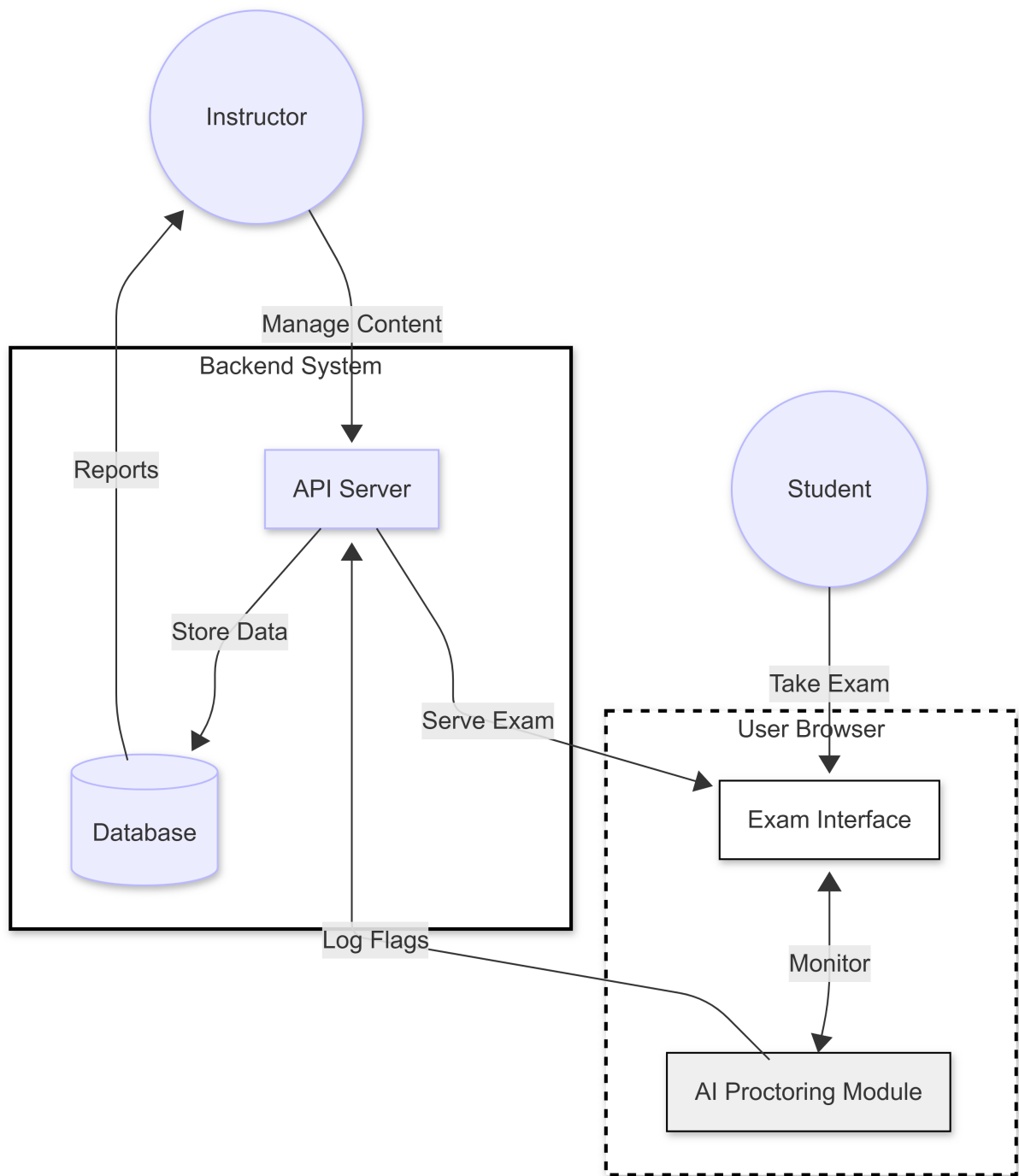


Figure 1.1: Entire work-flow

2 Overall Description

2.1 Product Perspective

”Focus Flow” is a replacement for fragmented tools like Google Forms combined with separate coding compilers. Unlike enterprise services that are high-cost and privacy-invasive, this project uses a tailored MERN-stack application with client-side ML to ensure privacy and efficiency.

2.2 User Classes and Characteristics

”Focus Flow” has basically 3 types of stakeholders. Refer Figure 2.1

- Students
 - Primary users who take exams and review learning progress.
- Instructors/Faculty
 - Responsible for designing assessments and reviewing integrity reports.
- Educational Institutions
 - Organizations deploying the system.

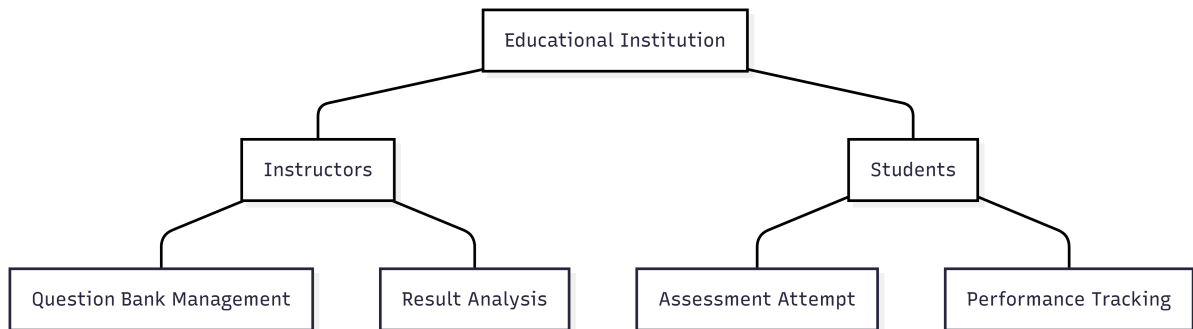


Figure 2.1: User hierarchy

2.3 Product Functions

”Focus Flow” integrates several key functions to ensure a seamless assessment process.

Before using the main functions, users must be authenticated securely using Sessions and Cookies (Passport.js).

Key Functions include:

- **Exam Builder:** Instructors create dynamic exams with support for rich text and images. Includes bulk-upload features.
- **Coding Sandbox:** A built-in Monaco Editor allowing students to write/compile code within the browser with syntax highlighting.
- **AI Proctoring:** A client-side ML module (MediaPipe) monitors behavior (looking away, leaving tab) and generates ”Flag Events”.
- **Analytics:** Post-exam visualization and comparative graphs for students.

2.4 Operating Environment

The website will operate in any modern web browser (Chrome, Edge, Firefox) on various Operating Systems (Windows, Mac, Linux). It requires a webcam for the proctoring features.

2.5 Design

The application follows a responsive web design philosophy.

Student Activities:

- Login/Register
- Attempt Exam (MCQ + Coding)
- View Results/Analytics
- Refer Figure 2.2

Instructor Activities:

- Create/Manage Exams
- Manage Question Banks
- Review Results and Proctoring Logs
- Refer Figure 2.3

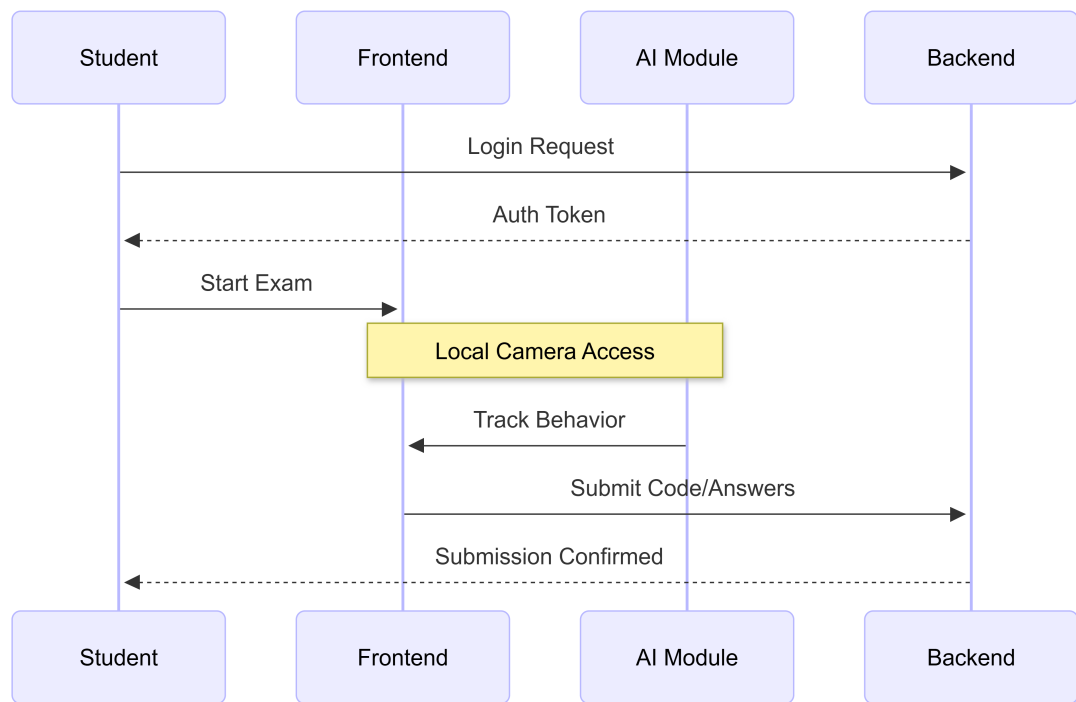


Figure 2.2: Student Activities



Figure 2.3: Instructor activities

3 System Features

"Focus Flow" is an AI-Proctored Exam & Learning Management System. The main art of this product is to ensure exam integrity without invading privacy.

3.1 Description and Priority

The features with priority up to down -

1. **User Authentication:** Secure login is foundational (Passport.js).
2. **Exam Delivery:** The ability to serve questions and accept answers.
3. **AI Proctoring:** The core differentiator; monitoring gaze and tab activity.
4. **Coding Sandbox:** Essential for technical assessments (Monaco Editor).
5. **Result Generation:** Automating grading and feedback.

3.2 Functional Requirements

The "Focus Flow" website is being built on the MERN stack.

Front-End:

- React.js (Dynamic UI, State management)
- Bootstrap 5 (Responsive Design)
- MediaPipe (Client-side Machine Learning for proctoring)

Back-End:

- Node.js (Runtime environment)
- Express.js (RESTful API, Routing)

Database:

- MongoDB Atlas (NoSQL database for flexible documents)

4 Other Nonfunctional Requirements

4.1 Performance Requirements

"Focus Flow" is built as a responsive web application to ensure fast load times and a seamless user experience. The AI model runs client-side to reduce server bandwidth requirements and latency.

4.2 Security Requirements

- **Privacy-First Proctoring:** Video feeds are processed locally on the client and never sent to a server. Only "Flag Events" are stored. Refer Figure 4.1
- **Authentication:** Secure session management using Passport.js.
- **Data Integrity:** Secure storage of exam papers and submission logs.

4.3 Software Quality Attributes

- **Reliability:** The system must handle high concurrency during exam times.
- **Usability:** An intuitive interface for both students and instructors.
- **Maintainability:** Modular code structure using MERN stack best practices.

4.4 Business Rules

"Focus Flow" aims to digitize the examination process while maintaining the sanctity of assessments. It reduces paper usage and logistical overhead associated with physical examinations.

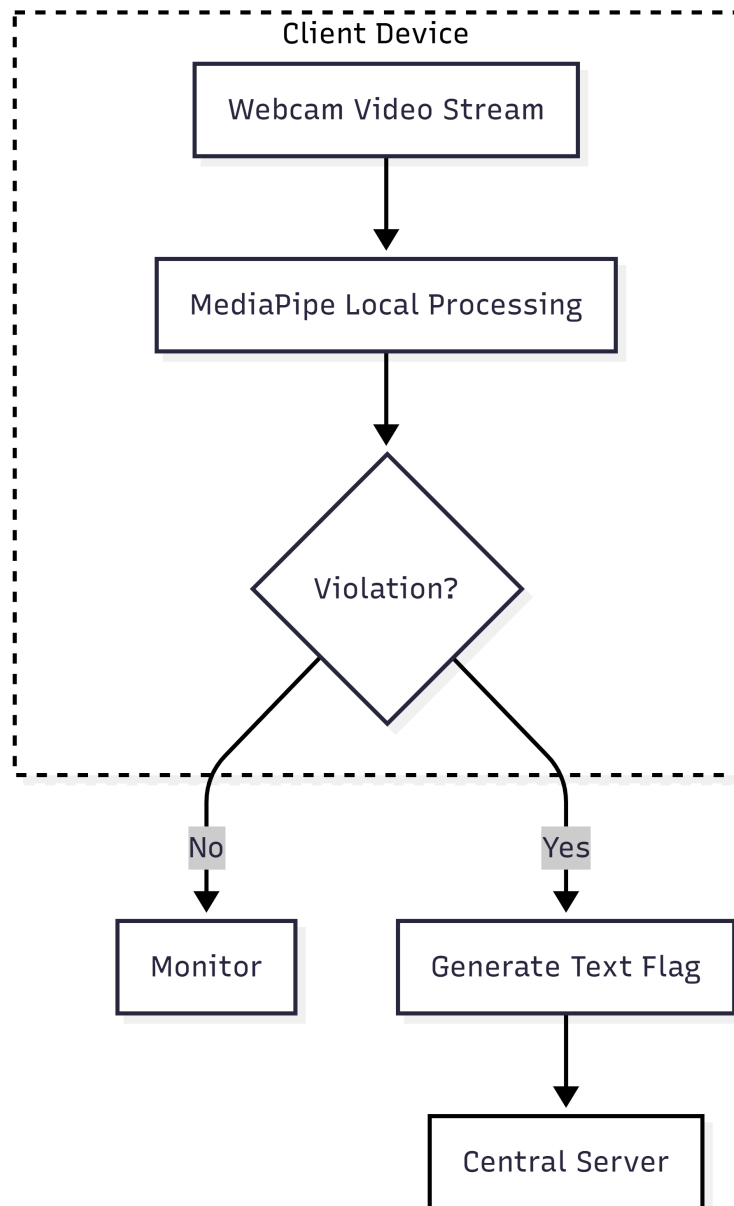


Figure 4.1: Privacy-First Proctoring

5 Other Requirements

The project requires continuous testing of the AI module to ensure it does not produce false positives during proctoring. The system also requires a functional webcam on the client device.

References

1. Google LLC, *MediaPipe Documentation*:
<https://developers.google.com/mediapipe> Accessed on 21/12/2025
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Appendix A: Glossary

AI Proctoring The use of automated machine learning algorithms to monitor student behavior during an exam to ensure academic integrity.

Flag Event A digital log generated by the AI module when a potential violation is detected, such as "looking away" or "leaving the tab".

MERN Stack A collection of JavaScript-based technologies used to build the application: MongoDB, Express.js, React.js, and Node.js.

MediaPipe An open-source framework by Google used in Focus Flow for client-side, privacy-first ML processing of video streams.

Passport.js Authentication middleware for Node.js used in Focus Flow to manage secure user sessions and cookies.